TD1

Exercise 1

Consider a smart light bulb that can be controlled using a smartphone and measures its own power consumption.

a) Define the Internet of Things (IoT) in one sentence, using your own words.
b) Identify the sensor(s) and actuator(s) in this smart light bulb.
c) Explain the difference between a connected object and a "smart" object.
d) List the four hardware building blocks found in almost every connected device.

Exercise 2

An urban video surveillance system connects 500 cameras to a central platform.

a) For each of the eight characteristics studied in class (connectivity, identification, sensing, intelligence, heterogeneity, scalability, autonomy, and security), provide a concrete example from this system.
b) Does this system need to support scalability? Justify your answer.
c) Suggest two resource constraints that would apply to a camera powered by solar energy and a battery.

Exercise 3

Classify each of the following scenarios as M2M, IoT, or both, and justify your answer in one sentence.

a) A gas meter transmits its reading once a month to a dedicated server operated by the utility provider, using a proprietary cellular connection.
b) A fleet of shared bicycles sends the GPS location of each bicycle to an open platform used by several applications (city authorities, the operator, and users).
c) A vending machine alerts its supplier when it needs to be restocked.
d) A city combines data from several types of sensors (traffic, pollution, and parking) to optimize public street lighting.

Exercise 4

Choose two domains from the following list: smart homes, connected healthcare, smart agriculture, Industry 4.0, and smart cities.

a) Describe a concrete use-case scenario for each selected domain in at least three sentences.
b) Identify the sensors and/or actuators required for each scenario.
c) Identify one dominant constraint for each scenario (communication range, energy consumption, latency, reliability, or cost) and justify your choice.

Exercise 5

a) Describe, in three steps, how a temperature measurement travels through a wireless sensor network (WSN) to reach a sink node.

b) Explain the difference between the SaaS, PaaS, and IaaS service models, providing one example of each.
c) Define the five Vs of Big Data and provide one IoT-related example for each V.
d) What is an embedded system? Identify two constraints that distinguish it from a conventional computer.
e) An airbag must respond within a few milliseconds. What type of timing constraint does this represent? Justify your answer.

Exercise 6

A connected wristband is being designed to measure heart rate and encrypt the data before transmitting it.

a) Explain the difference between top-down and bottom-up design approaches.
b) Propose a hardware/software partitioning for this wristband. Which functions should be implemented in hardware, and which should be implemented in software? Justify your choices.
c) Why is encryption often delegated to a dedicated hardware component rather than implemented solely as a software program?

Exercise 7

a) Identify three major IoT challenges and describe one concrete consequence of failing to address each challenge.
b) Briefly explain how the Mirai botnet attack (2016) worked and suggest one simple security measure that could have limited its impact.
c) A weather station operates using a 3,000 mAh battery. Its average current consumption is 0.15 mA.

- Calculate its theoretical operating time in hours.
- Convert this operating time into years, assuming that one year is approximately 8,760 hours.
- Comment on the result.